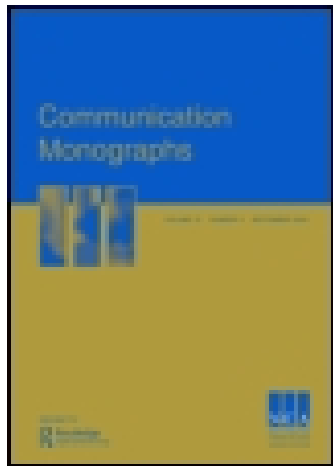




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DECISION EMERGENCE: PHASES IN GROUP DECISION-MAKING

B. AUBREY FISHER

THE study of small groups long has been a major interest in the behavioral sciences. Since the famed Hawthorne studies,¹ researchers fruitfully have probed the socio-emotional dimension of groups attempting to explain, among other things, the "Hawthorne effect." Unfortunately, in-depth research of task behavior has not kept pace. McGrath and Altman, following an intensive study of small group research, called for "more attention to analysis of the behavior of groups—their interaction—as opposed to the products or outcomes of that behavior."² For those whose primary interest is the process of speech communication, the need for careful analysis of the group interaction process is acute.

This study is concerned with the group task of decision-making—typically, but not necessarily, problem-solving. Without sufficient insight into the nature of a group's interaction process, the typical approach³ to group decision-making has followed John Dewey's⁴ pat-

tern of reflective thinking for individuals. However, Kelley and Thibaut,⁵ among others, have warned against substituting an individual process for the group process.

There have been few attempts to test theoretical conceptions of group task behaviors systematically. Among the most notable of such conceptions are Bales and Strodtbeck's⁶ three-phase progression of group development, Bennis and Shepard's⁷ four levels of work, and Scheidel and Crowell's⁸ spiral process of idea development. Each of these studies, however, possesses serious limitations. Bales and Strodtbeck, for example, reported that groups concentrated on different kinds of problems across time but they did not discuss the interaction process which the groups used to solve those problems. Bennis and Shepard's study was concerned primarily with delineating those socio-emotional variables which obstruct efficient task completion. Only Scheidel and Crowell observed the verbal behavior related specifically to task accomplishment—notably decision-making. Unfortunately, these

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¹For a comprehensive account of the Hawthorne studies, See F. J. Roethlisberger and William J. Dickson, *Management and the Worker* (Cambridge, 1946).

²Joseph E. McGrath and Irwin Altman, *Small Group Research: A Synthesis and Critique of the Field* (New York, 1966), p. 3.

³See, for example, J. F. Dashiell, "Experimental Studies of the Influence of Social Situations on the Behavior of Individual Human Adults," *A Handbook of Social Psychology*, ed. C. Murchison (Worcester, 1935), p. 1131; James G. March and Herbert A. Simon, *Organizations* (New York, 1958), p. 179; and Abraham Zelenz-nik and David Moment, *The Dynamics of Interpersonal Behavior* (New York, 1964), p. 152.

⁴See John Dewey, *How We Think* (Boston,

1910) and *Logic: The Theory of Inquiry* (New York, 1938).

⁵Harold H. Kelley and John W. Thibaut, "Experimental Studies of Group Problem Solving and Process," *Handbook of Social Psychology*, ed. Gardner Lindzey (Reading, 1954), II, 738-741.

⁶Robert F. Bales and Fred L. Strodtbeck, "Phases in Group Problem-Solving," *Journal of Abnormal and Social Psychology*, XLVI (1951), 485-495.

⁷Warren G. Bennis and Herbert A. Shepard, "A Theory of Group Development," *Human Relations*, IX (1956), 415-437.

⁸Thomas M. Scheidel and Laura Crowell, "Idea Development in Small Discussion Groups," *QJS*, L (1964), 40-45.

authors did not report observing aspects of the time dimension which McGrath and Altman consider "absolutely crucial for understanding small group phenomena."⁹

The purpose of the present investigation, then, was to discover the nature of the interaction process across time leading to group consensus on decision-making tasks. A pilot study involving intensive observation of classroom and non-classroom group discussions revealed several general characteristics of task behavior consistent among all groups. First, an accumulation of verbal cues, reinforcing or resisting the group's tendency to arrive at decisions, determine the ultimate but gradual emergence of those decisions. Second, this pattern of verbal cues operates independently from the group's social dimension or level of cohesiveness. Third, a linear problem-solving agenda suitable for individual decision-making was not typically evident. Groups who consciously elected to use such a system typically reached decisions despite, rather than because of, that system.

These group characteristics from the pilot study suggested that group members perceive several alternatives for accomplishing their tasks. As discussions progress, these alternatives are debated, refined, accepted, changed, rejected, etc., until the final task product evolves from their deliberations—largely through a "method of residues" or elimination of available alternatives. Although they did not discuss the interaction process involved, Kelley and Thibaut¹⁰ believed that group decision-making involved a process of choosing among alternatives. The present investigation of the interaction process characteristic of decision-making small groups proceeded from these preliminary observations.

METHOD

Selection of Groups. After careful examination of available groups and the purposes of the investigation, the following criteria were used to select groups for observation:

1. The group's goal is to achieve consensus on decisions.
2. The group's interaction is primarily verbal.
3. The group operates in a developing or emergent context.
4. The entire history of the group is available for study.
5. A written record of the group's task performance is available.
6. The group successfully accomplished its task.
7. The group is not formed within a classroom context.

Despite these common characteristics, the ten selected groups displayed a variety of dissimilarities. They varied in size from four to twelve members, in duration from twenty-five minutes to over thirty hours, in composition from all males to all females, in age from teens to the sixties. These differences imply a diversity of social structures, role relationships, leadership methods, and decision quality, among others. The maturity and attitudes of the members, their experience and expertness with using the group method, and so forth, also varied. The present study purposely did not attempt to control variables in the social dimension. Since the purpose was to discover patterns of verbal task behavior regardless of socio-emotional structures and individual differences among participants, the diversity of social dimensions among the groups was vital.

Content Analysis: The Category System. Audiotapes of the verbal interaction of each group's entire history were

⁹ McGrath and Altman, p. 73.

¹⁰ Kelley and Thibaut, p. 744.

transcribed and submitted to content analysis using an original category system expressly designed to exclude forces of the socio-emotional dimension apparent in the interaction. Any interaction considered to be serving purely a procedural or socializing function, i.e., non-task function, was coded into an "et cetera" category and excluded from the data.

The concept of *decision proposal* is central to the operation of the category system. Identifying the specific decision alternative being considered by the group anchors verbal interaction to the subject matter of the discussion. Verbal interaction, as defined by this study, then, processes information pertinent to a specific decision proposal and leads to its eventual acceptance or rejection by the group. The difference between task-function categories is the difference between functions performed on the decision proposal. If one assumes that a group reaches consensus via interaction, then the observed patterns of interaction functioning on a group's perception of decision proposals should reveal how groups use interaction to achieve consensus on those decision proposals. The categories, taken together, comprise a list of all possible task functions which verbal behavior can perform.

The *unit* of interaction is defined as a given instance of the performance of a task function on a decision proposal regardless of its length or intensity. The distinction between two units of interaction, therefore, is the interruption of one individual's contribution by another individual or the crossing of function categories within an individual's contribution. Defining the unit as a function attempt rather than a thought-unit, then, results in dispersing the data from interaction analysis among group members. In general, "n" units may be interpreted as nearly "n" individual comments con-

taining "n" different attempts to direct the group's perception of a decision proposal.

The category system described each unit of interaction along several dimensions, each of which contains mutually exclusive and exhaustive categories.

Dimension One:

- A—Asserted
- S—Seeking

Dimension Two:

- 1—Interpretation
- 2—Substantiation
- 3—Clarification
- 4—Modification
- 5—Summary
- 6—Agreement

Dimension Three:

(applicable to "1" and "2" categories only)

- f—Favorable toward the proposal
- u—Unfavorable toward the proposal
- a—Ambiguous toward the proposal

Additional Symbols:

- O_n—Origin of a decision proposal
- (D_n)—Reintroduction of a proposal

Hence, a unit coded "A210₃" would indicate an asserted contribution introducing into the discussion and favorably substantiating decision proposal number three. Agreement among minimally trained coders using the system exceeded .86 in several checks for reliability of the category system.

Content Analysis: Static and Contiguity Analyses. After the transcripts of verbal interaction were coded into units of the category system, each group was arbitrarily divided into five-minute segments. Each coded comment in each segment for each group was then transferred to a contiguity table for analysis, adopting a procedure used by Scheidel and Crowell.¹¹ The "contiguity analysis," utilizing the units as both rows and columns in the table, identified each contiguous pair of function units. That is,

¹¹ Scheidel and Crowell, pp. 142-143.

the rows of the table indicated the antecedent unit of each pair; the column indicated the subsequent. Therefore, each cell in the table included the frequency with which the column-unit immediately followed the row-unit during that five-minute segment of time. Since an inspection of the data revealed nearly all the units to be assertions, the contiguity table did not distinguish between asserted and seeking comments.

A "static analysis," i.e., the total number of units in each category in each segment, is available from the contiguity table by using the marginal totals of the rows—the antecedent units. But the contiguity analysis of this study provides, in addition, a key to group interaction not available with most category systems. Units quantifying contiguous pairs of function attempts typically indexed interaction between pairs of members when the system was applied to the group transcripts. Weick¹² considers the "interact," as well as the "act," an essential unit of collective structure. McGrath and Altman apparently concur in stating, "there are strong conceptual reasons for preferring dyadic or polyadic units (interacts of two or more group members)."¹³ The present study attempts to consider both "acts" and "interacts" within its scope of interaction analysis. Thus, static analysis of the data provides units of "acts," and contiguity analysis of the data provides units of "interacts."

Division into Phases. Since the pilot study indicated a changing pattern of verbal cues reinforcing or resisting the group's tendency to arrive at a decision, the data from the third dimension of the category system were observed for each five-minute segment of each group. A consistent four-phase pattern emerged from a study of the proportional fluctua-

tion of interaction units favorable, unfavorable, and ambiguous toward the decision proposals. In the early stages of the discussion, both favorable attitudes and ambiguous attitudes were expressed with greater frequency than unfavorable attitudes. Then, as the proportion of ambiguous units declined, unfavorable units increased to surpass the proportional frequency of ambiguous ones. In a third phase, unfavorable units declined while the proportional frequency of ambiguous units rose above the frequency of unfavorable ones. In the final phase, favorable units rose sharply, and both ambiguous and unfavorable units declined sharply.

This four-stage pattern appeared in each of the ten groups studied with slightly varying degrees of distinction between phases. In all ten groups, the distinctions between the first and second and between the third and fourth phases were clearly evident. The distinction between the second and third phases was not so clear-cut in all ten groups. Distinguishing the four phases, then, does not imply the presence of clearly discrete phases of progression. Rather, the phasic progression reflects a continuous and gradual change of interaction patterns.

Quantitative Analysis. Each of the ten groups was separated into four phases on the basis of the variations in the third dimension (favorability). The data from all ten groups were then collapsed into the four phases and transferred to contiguity and static analysis tables utilizing the four phases as the rows and the various combinations of units and pairs of units as the columns. Three tables for contiguity analysis and three tables for static analysis were formed by collapsing all the units into each of the three dimensions. Thus, 109 categories of interaction appeared as column headings in the six tables.

The Pearson chi-square test of associ-

¹² Karl E. Weick, *The Social Psychology of Organizing* (Reading, 1969), pp. 45-48.

¹³ McGrath and Altman, p. 74.

TABLE 1
RESULTS OF CONTENT ANALYSIS FOR PHASE ONE

		Subsequent Units							
		1f	1u	1a	2f	2u	2a	3	6
Antecedent Units	1f	19	12	14	16	4	10	21	14
	1u	7	12	11	16	6	5	8	19
	1a	16	13	29	6	9	20	15	18
	2f	17	11	5	25	16	12	25	17
	2u	6	6	10	11	8	7	18	9
	2a	3	6	13	9	7	13	22	17
	3	18	10	20	27	17	16	154	24
	6	21	13	16	14	9	11	25	XX

Total pairs of units categorized = 1008.

TABLE 2
RESULTS OF CONTENT ANALYSIS FOR PHASE TWO

		Subsequent Units							
		1f	1u	1a	2f	2u	2a	3	6
Antecedent Units	1f	37	18	13	10	9	6	10	21
	1u	16	22	4	12	7	5	10	18
	1a	11	8	10	9	6	3	13	7
	2f	13	15	10	34	20	12	19	23
	2u	7	10	5	28	17	12	7	6
	2a	5	3	2	9	10	9	13	3
	3	12	10	8	23	10	13	122	3
	6	18	8	7	23	9	1	4	XX

Total pairs of units categorized = 848.

TABLE 3
RESULTS OF CONTENT ANALYSIS FOR PHASE THREE

		Subsequent Units							
		1f	1u	1a	2f	2u	2a	3	6
Antecedent Units	1f	53	28	35	35	14	14	25	19
	1u	20	12	18	20	5	9	9	4
	1a	39	11	42	24	11	21	19	8
	2f	36	19	25	47	28	19	26	34
	2u	13	7	9	31	6	8	9	4
	2a	9	5	22	31	7	24	34	11
	3	33	9	15	26	7	10	170	6
	6	16	8	8	22	7	9	7	XX

Total pairs of units categorized = 1312.

TABLE 4
RESULTS OF CONTENT ANALYSIS FOR PHASE FOUR

		Subsequent Units							
		1f	1u	1a	2f	2u	2a	3	6
Antecedent Units	1f	72	7	13	26	3	4	20	23
	1u	6	2	1	5	0	2	1	0
	1a	15	1	7	8	1	1	3	2
	2f	23	2	8	44	4	9	20	15
	2u	3	0	1	10	1	0	2	0
	2a	3	0	2	10	0	5	2	1
	3	23	5	2	9	4	1	70	7
	6	18	0	4	11	1	1	5	XX

Total pairs of units categorized = 549.

ation¹⁴ was selected to determine whether the four phases were independent of the categories of content analysis. So that the tests for independence would indicate which categories of interaction distinguished between which phases, each of the 109 categories from static and contiguity analysis was systematically tested for every possible combination of two phases by replicating two-by-two contingency tables. The frequency of each category was compared against the combined frequencies of all other categories in phases one and two, one and three, one and four, two and three, two and four, and three and four. In this fashion each category was compared across all four phases using six contingency tables for each category-unit. Each rejected null hypothesis, then, aids in distinguishing more precisely, on the basis of the category system, the interaction patterns of the four phases.

RESULTS

The tables reveal some categories to be barren of information. In addition to the category of "seeking," the number of units coded into the "modification" (4)¹⁵ and "summary" (5) categories were too few to warrant attention. Thus, all such units, including all interacts which contained either as the antecedent or subsequent unit, were eliminated from further consideration. Of the 3,717 units remaining, phase one contained 1,008 and phase two contained 848 (see Tables 1 and 2). Phase three appeared to be the longest phase typically with 1,312 units (see Table 3). Conversely, phase four was typically the shortest with only 549 units (see Table 4). The table for this

final phase is the only one with empty cells. The extremely low frequency of interaction units expressing unfavorable attitudes toward decision proposals (1u, 2u) resulted in these null cells.

Of the 654 null hypotheses of independence of category-units and phases, 217 were rejected with results beyond the .05 level of significance. Table 5 shows the units for which significant chi-square results were obtained between each pair of phases. Of the 217 rejected hypotheses, 40 are labelled questionable because of an expected frequency less than ten.¹⁶ Nearly all of these 40 are the result of low frequencies of unfavorable (u, 1u, 2u) or ambiguous (a, 1a, 2a) category units in phase four.

DISCUSSION

The results of the chi-square tests indicate that the four phases can be differentiated on the basis of an interaction pattern characteristic of each phase. The following is a description of the distinctive interaction pattern found for each phase:

Phase One. Significantly more clarification (3) and agreement (6) occurred in phase one than in any of the other three phases. It appears that a group's early problems of socializing and developing a socio-emotional climate conducive to task accomplishment affected the task interaction patterns in the early phase. As each member was unaware of his initial social position in the group, he did not quickly or strongly assert himself or his opinions. Consequently, he made assertions tentatively in order to "test" the group.

The categories of interaction antecedent to agreement (6) substantiates

¹⁴A complete description of this test and its applications is found in William L. Hays, *Statistics for Psychologists* (New York, 1963), pp. 589-603.

¹⁵Letters and numbers in parentheses indicate the coding symbol used to delineate that function category within the category system.

¹⁶Hays, p. 597, indicates that a conservative rule of thumb for contingency tables with a single degree of freedom is a minimum expected frequency of ten.

TABLE 5
SIGNIFICANT DIFFERENCES AMONG PHASES BASED ON UNITS OF INTERACTION

Unit	One	Two	Phases Three	Four	Unit	One	Two	Phases Three	Four
(Reinforced Agreement)					(Expressed Attitudes)				
1f/1f		>1	>1	>1,2,3	1f		>1	>1	>1,2,3
1f/2f			>2	>1,2,3	1u	>4	>3,4	>4	
1f/6				>1,3	1a	>2,4		>2,4	
2f/1f				>1,2	2f		>1	>1	>1,2,3
2f/2f				>1,2*,3	2u	>4	>1,3,4	>4	
6/1f				>3	2a	>2,4		>2,4	
6/2f	>1				f		>1	>1	>1,2,3
f/f			>1	>1,2,3	u	>3,4	>1,3,4	>4	
f/6				>1,3	a	>2,4		>2,4	
6/f		>3		>3	(Tenacity of Attitudes)				
(Reinforced Disagreement)					1/1			>1	>1,2
1u/1u		>1,3,4			1/2			>1,2	
1u/2u		>4*			1/6	>3	>3		>3
1u/6	>3,4*		>3*,4*		2/1			>1,2,4	
2u/2u		>1,3,4*			2/2		>1	>1	>1
2u/6	>4*	>4*			6/1	>3			
6/1u	>4	>4*			1			>1,2	>1,2
u/u	>3,4	>1,3,4			2		>1	>1,4	
u/6	>3,4	>3,4			(Clarification)				
6/u	>3,4	>4*	>4*		1f/3				>2,3
(Reinforced Ambiguous Disagreement)					1u/3		>4*		
1a/1a	>2,4		>2,3		2f/3			>4	
1a/2a	>2,4*		>2,4*		2u/3	>3,4*			
1a/6	>3,4*				2a/3	>4*	>4*	>4	
2a/1a	>2*		>2,4*		3/1f	>4			>2
2a/6	>2*,4*				3/1a				>1*
6/1a	>3				3/2u	>3			
6/2a	>2*				3/2a	>4	>4*		
a/a	>2,4		>2,3		3/6	>2,3	>4*		
a/6	>2,3,4				6/3	>2,3,4			
6/a	>2,3,4				3/2	>3,4	>3,4		
(Overt Conflict)					3	>2,3,4			
1u/1f		>1			6	>3,4	>3		
2f/1u		>4*	>4*		(Et cetera)				
2f/2u		>4*	>4		1u/1a			>2*,4*	
2u/2f		>1	>1		1a/1u	>4*			
f/u		>1,4	>4		2u/2a	>4*	>4*		
u/f		>1,4			2a/2u	>4*	>4*		
(Ambiguous Conflict)					u/a		>4	>4	>1
1f/1a			>1		a/u	>3,4	>4	>4	
1a/1f			>1,2						
1a/2f			>1						
2f/1a			>1	>1*					
2a/2f			>1,2						
f/a			>1,2						
a/f			>1,2	>1,2					

*. . . Questionable result ($ef < 10$).

this explanation. For instance, members agreed with significantly more ambiguous comments (a/6) and ambiguous interpretations (1a/6) in the first phase. The frequency of ambiguous comments following agreement (6/a) was also significantly greatest in the first phase. In-

terestingly, members also agreed more often with unfavorable comments (u/6) in the first phase than in either of the latter two phases. And unfavorable comments followed agreement (6/u) more often in the first phase than in either of the last two.

Indicative of the tendency to agree to nearly everything in the early phase is the discovery that agreement followed clarification (3/6) more often in phase one than in either of the next two phases. Clarification also followed agreement (6/3) most often in the first phase. Agreement may function in phase one, then, not so much to reinforce opinions and beliefs as to encourage social facilitation or to avoid disrupting the developing social climate.

Group members in phase one tended to search tentatively for ideas and directions to aid their decision-making efforts. Since the members were unaware of the direction the group would ultimately pursue, fewer favorable comments (f, 1f, 2f) appear in this phase than in any other, and more unfavorable comments appear than in either phase three or four. The first phase also contains fewer unfavorable comments reinforced by unfavorable comments (u/u) than either of the next two phases.

Favorable comments were least reinforced in phase one, i.e., favorable interpretation followed by favorable interpretation (1f/1f), perhaps because of the members' tentative opinions, typical of task communication in a developing social structure. Fewer favorable comments followed other favorable comments (f/f) in phase one than in either phase three or phase four. Tentative opinions, of course, do not demand substantiation. Thus, significantly fewer substantiation units (2) appear in phase one than in either of the final two phases.

Attitude reinforcement, for the purposes of this study, is defined as an antecedent unit followed by either agreement or a unit expressing the same attitude toward the decision proposal. Indicative of the tentative nature of early task communication is the random pattern of attitude reinforcement. That is, group members do not reinforce predominantly one

attitude toward decision proposals. Rather, they reinforce significantly more ambiguous comments (a/a) than in phases two or four. On the other hand, when compared with either the third or fourth phase, phase one contains more unfavorable comments reinforced by agreement (u/6) and other unfavorable comments (u/u).

Unexpectedly, members also reinforced favorable units less often in the earliest phase. For example, phase one contains significantly fewer favorable interpretation comments reinforced by other favorable interpretation (1f/1f). When compared to either phase three or phase four, the first phase contains less reinforcement of favorable comments with favorable comments (f/f). Several factors can account for these data. First, each of the other phases contains more favorable comments (f, 1f, 2f) than does the first, undoubtedly because the group progressively approaches consensus. Second, since reinforcement is sporadic rather than systematic, the first phase includes proportionately more reinforcement of unfavorable and ambiguous comments. Hence, there is proportionately less reinforcement of favorable comments. Such explanations, while justified statistically, add little to our understanding of group task interaction.

The function of ambiguous units provides the key to understanding the first phase. More ambiguous comments (a, 1a, 2a) are contained in phase one than in any other phase except the third, which is also characterized by ambiguity. Many ambiguous comments in the first phase probably reflect favorable attitudes toward decision proposals. That is, since favorable attitudes assert themselves with increasing intensity throughout the four phases, they must appear in phase one in rudimentary form. Since favorable comments are observed relatively infrequently in the first-phase, they

are probably expressed as ambiguous attitudes. As the issues become clarified and as the social climate becomes more conducive to the honest statement of one's true position, those opinions appear as favorable units.

Characteristic of the first phase, then, is getting acquainted, clarifying, and tentatively expressing attitudes. Bales and Strodtbeck¹⁷ described the initial stage of their three-phase group development as emphasizing problems of orientation. Schutz¹⁸ also characterizes his initial stage of "inclusion" as a period of members orienting themselves to the group atmosphere and testing ambiguous "goblet issues" of questionable relevance. The task activity of Tuckman's¹⁹ first phase, "forming," is also characterized by orientation. The present data reveal characteristics consistent with these descriptions of group development. Therefore, the first phase of this decision-making model will be called the *orientation phase*.

Phase Two. The period in a group's history of task behavior following the orientation phase is characterized by dispute. For example, more unfavorable (u) and unfavorable substantiation (2u) units are contained in the second phase than in any other phase, and more unfavorable interpretation (1u) units appear in phase two than in the two succeeding phases.

In the orientation phase, members tentatively formulate opinions. By the second phase, they are much more definite. Along with the highest level of unfavorable comments in phase two, the number of favorable (f), favorable interpretation (1f), and favorable substantia-

tion (2f) units increased significantly from the orientation phase. But ambiguous (a), ambiguous interpretation (1a), and ambiguous substantiation (2a) units are significantly fewer than in phases one or three. The members in the second phase are aware of the direction the groups' task behavior is heading, of the relevant decision proposals which are emerging from the group deliberations. Thus, members typically express either a favorable or an unfavorable attitude toward those decision proposals rather than an attitude of ambiguity.

With polarization of attitudes comes disagreement. Comparing the orientation phase with the second phase reveals a significant increase in the frequency of unfavorable interpretation followed by favorable interpretation (1u/1f) and unfavorable substantiation followed by favorable substantiation (2u/2f). Ideational conflict is further evidenced by more unfavorable comments followed by favorable comments (u/f) than in either the first or the last phase.

Characteristic of second phase conflict is the greater tendency to disagree with unfavorable than with favorable comments. That is, unfavorable units are resisted in phase two by subsequent favorable units (1u/1f, 2u/2f). No unquestioned significant difference was found between phase two and any other phase in the frequencies of favorable interpretation and favorable substantiation followed by any unfavorable comment (1f/1u, 1f/2u, 2f/1u, 2f/2u). However, collapsing interpretation and substantiation indicates a significantly greater number of favorable units followed by unfavorable units (f/u) in phase two.

Not only are comments in the second phase less ambiguous, they are also expressed more tenaciously. Phase two is characterized by attempts to persuade dissenting members. It contains a smaller proportion of interpretation units (1)

¹⁷ Bales and Strodtbeck.

¹⁸ William C. Schutz, "Interpersonal Underworld," *Harvard Business Review*, XXXVI (1958), 123-135.

¹⁹ Bruce W. Tuckman, "Developmental Sequence in Small Groups," *Psychological Bulletin*, LXIII (1965), 384-399.

than either phase three or phase four. Argument form, i.e., premises and data leading to a conclusion, is the appropriate form for debate, and the significantly greater number of substantiation units (2, 2/2) indicates the argumentative quality of this phase, unlike orientation.

The reinforcement schedule of the second phase leads to the inference that coalitions have been formed. While reinforcement was random in the orientation phase, it is bimodally distributed in the second phase. The frequency of favorable interpretation comments following favorable interpretation comments (1f/1f), for example, increased significantly from the orientation phase. And only the fourth phase contains significantly more favorable comments followed by other favorable comments (f/f). But unfavorable comments reinforced by other unfavorable comments (u/u) and unfavorable interpretation reinforced by unfavorable interpretation (1u/1u) and unfavorable substantiation units reinforced by unfavorable substantiation (2u/2u) occurred with significantly greater frequency in the second phase than in either phase three or four. On the other hand, ambiguous units (a, 1a, 2a) appeared less frequently in phase two and were subsequently not reinforced.

The reinforcement pattern indicates the presence of coalitions formed from ideational polarization. Because of the nature of the category system and the unit measuring interaction, whenever a dyadic unit (e.g., 1f/1f, 1u/1u, etc.) appears, it typically indicates an interaction between two members. The dyadic units most prevalent in phase two imply coalitions of individuals. In other words, two coalitions are present in the second phase—one favoring and one opposing those decision proposals which ultimately achieve consensus. Of course, these coalitions are relevant only to the

extent that they involve attitudes toward decision proposals. In no way do the data from this study preclude the presence of other coalitions. Coalitions that are idiosyncratic to a particular group or that characterize unsuccessful groups or that result from socio-emotional pressures are not necessarily revealed by these data.

The second phase, therefore, is characterized by dissent. Polarization of attitudes points to this conclusion. The rise of dyadic interactions of favorable and unfavorable units is indicative of dissent, as is the increase of substantiation and substantiation dyads. Because of these distinguishing characteristics, phase two could be called the phase of controversy, argument, or conflict. The most descriptive of these terms is conflict—hence, the *conflict phase*.

Phase Three. Conflict and argument dissipate during phase three. For example, unfavorable substantiation (2u) and interpretation (1u) are significantly fewer than in the conflict phase. And phase three contains fewer unfavorable units (u) than either of the earlier phases. Consequently, unfavorable opinions, liberally reinforced in the conflict phase, receive less reinforcement (1u/1u, 2u/2u, u/u) in the third phase.

But residues of conflict remain, indicating that the reduction of conflict is gradual. More unfavorable substantiation followed by favorable substantiation (2u/2f), for example, appears in phase three than during orientation. And the argument form, typical of ideational conflicts, appears more often in the third phase than during orientation. That is, more dyadic interactions of substantiation (2/2) appear in phase three. Perhaps more importantly, phase three contains more interpretation followed by interpretation (1/1) and favorable comments followed by other favorable comments (f/f). While conflict may yet

exist in the third phase, the members do not defend unfavorable attitudes so tenaciously against persuasive attempts.

Conflict, denoted by the bimodal distribution of favorable and unfavorable units, is lower in the third phase than in any other except for the final phase. Only when associated with the fourth phase is phase three found to contain more unfavorable units (u , $1u$, $2u$), fewer favorable units (f , $1f$), less positive reinforcement of favorable attitudes (f/f), and more negative reinforcement (f/u). Thus, the level of conflict in phase three is significantly lower than in either preceding phase but somewhat higher than in the final phase.

The hallmark of the third phase is the recurrence of ambiguity. More ambiguous units (a , $1a$, $2a$) appear in phase three than in either the conflict phase or the fourth phase. As in the orientation phase, ambiguous units are reinforced in significantly greater proportion than in phases two or four; i.e., ambiguous interpretation followed by ambiguous interpretation ($1a/1a$), ambiguous substantiation followed by ambiguous interpretation ($2a/1a$), and ambiguous interpretation followed by ambiguous substantiation ($1a/2a$). Thus, ambiguity, prominent in the orientation phase, declines in the conflict phase, and reasserts itself in the third phase.

But while the *level* of ambiguity and reinforcement of ambiguity is not significantly different from the orientation phase, the *function* performed by ambiguity in phase three is. The discussion above of the orientation phase suggested that ambiguity served as the initial expression of tentatively favorable attitudes. This explanation of ambiguity is not appropriate to the third phase. For example, phase three contains fewer reciprocal reinforcement patterns ($a/6$ and $6/a$, $1a/6$ and $6/1a$); groups are no longer searching for attitude di-

rection. This direction was plotted in the orientation phase and debated in the conflict phase. In the third phase, task direction is no longer an issue. Hence, phase three contains significantly fewer dyadic interactions including clarification units ($3/6$, $6/3$, $2u/3$, $3/2$, $3/2u$), and significantly more favorable units (f) and favorable interpretation units ($1f$), and receive more positive reinforcement ($1f/1f$, f/f). Hence, ambiguity's role in phase three is not that of tentative attitude expression.

The key to the function of ambiguity in the third phase lies in associating phase three with the conflict phase. The earlier phase was typified by negatively reinforced attitudes bimodally distributed between favorable and unfavorable units. The preceding discussion has indicated that phase three contains fewer unfavorable units (u , $1u$, $2u$) and fewer positive reinforcement patterns of unfavorable attitudes (u/u , $1u/1u$, $2u/2u$). But ambiguous units (a , $1a$, $2a$) and positive reinforcement of ambiguous units ($1a/1a$, $2a/1a$, $1a/2a$) increased. Further comparison with the conflict phase reveals that the negative reinforcement of favorable units is fundamentally different in the third phase. That is, the third phase contains more favorable units followed by ambiguous units (f/a) and ambiguous units followed by favorable units (a/f , $1a/1f$, $2a/2f$). These same units also distinguish the third phase from the orientation phase, lending credence to the inference that ambiguity functions differently in phase three.

Ambiguity, then, functions in the third phase as a form of modified dissent. That is, the group members proceed to change their attitudes from disfavor to favor of the decision proposals through the mediation of ambiguity. Only the most naive observer of human behavior would speculate that attitude change would be abrupt, particularly

after these members had, during the conflict, publicly committed themselves to positions unfavorable to the decision proposals. Thus, dissent proceeds to assent via ambiguity. The conflict phase was characterized by dissent in the form of comments unfavorable to the decision proposals. The succeeding phase is characterized by dissipating dissent in the form of comments ambiguous toward the decision proposals.

The two coalitions present in the conflict phase also dissipate during the third phase. The dissenting coalition turns to ambiguous comments as a form of dissent. But since this dissipation of conflict is gradual and marked by ambiguity, the question concerning the point in time at which decisions are made must remain unanswered. In all likelihood, the group participants are as unaware as the observer of this point in time. In the absence of apparent conflict, decisions may appear to have been reached. But the ambiguous attitude, while not unfavorable, is not yet favorable. As disfavor dissipates to ambiguity, though, favorable attitudes increase concomitantly. This characterization of the third phase prompts the label *emergence phase*.

Phase Four. While groups tend to reach decisions during the emergence phase, members become increasingly aware of those decisions in the final phase. Argument is no longer important in phase four. Significantly more interpretation (1) is evident in the final phase than in either of the first two, and more favorable interpretation (1f) than in any preceding phase. Although the final phase contains more favorable substantiation (2f) than in any other phase, substantiation (2) is proportionately greater in the emergence phase. The argument form as a unit of interaction (2), then, characteristically favors the decision proposals in the final phase. That is, fewer

unfavorable substantiation units (2u) are contained in the final phase, and fewer ambiguous substantiation (2a) units appear in the final phase than in either phase one or three. Thus, the argument form appears in phase four but characteristically favors the consensus decisions.

Comments favoring the decision proposals are constantly reinforced in phase four. When compared with every other phase, the final phase contains more dyadic interactions reinforcing favorable attitudes (f/f, 1f/1f, 1f/2f, 2f/2f). Excluding the emergence phase, the final phase also contains more favorable substantiation units followed by favorable interpretation units reinforced by agreement (f/6, 1f/6). This preponderance of interaction favoring the decision proposals and reinforcing those favorable attitudes earmarks the final phase of group deliberations.

Dissent has all but vanished in phase four. The final phase contains the fewest unfavorable attitudes (u, 1u, 2u) and less positive reinforcement of unfavorable attitudes (u/u, u/6, 2u/6). Except for the orientation phase, the final phase contains the fewest dyadic interactions of negative reinforcement (f/u, 2f/1u, 2f/2u). But while the low level of conflict during orientation reflected a conscious avoidance of conflict, in the final phase it reflects unity of opinion among group members. Analysis of the interaction also reflects this difference between the first and fourth phases. That is, phase four contains a significantly greater number of units favoring decision proposals (f, 1f, 2f).

Ambiguous dissent, characterizing the emergence phase, also dissipates in the final phase. For example, phase four contains fewer ambiguous units (a) and less reinforcement of ambiguous attitudes (a/a, 1a/1a). Yet some modified dissent remains. The frequency of favor-

able units following ambiguous units (a/f), for example, is greater in the final phase when it is compared with either of the first two phases. Of course, the gradual dissipation of dissent indicates that some residues from the emergence phase would probably appear in the succeeding phase, particularly early in the latter phase. Then, too, the low frequency of favorable comments in the orientation phase and the low frequency of ambiguous comments in the conflict phase make these particular interphase comparisons less meaningful. With the very high frequency of favorable comments and extremely low frequency of ambiguous and unfavorable comments in the final phase, one can safely conclude that the dissipation of dissent—modified or unmodified—is virtually complete.

This characteristic pattern of predominantly favorable attitudes consistently receiving positive reinforcement distinguishes the final phase of the group histories. Pervading the final stage is a spirit of unity. All members are in agreement and strive to show that agreement through positive reinforcement. Dissenting opinions, either unfavorable or ambiguous, have all but vanished. If decisions emerged during the emergence phase, they are reinforced during this final phase. Hence, the final phase bears the name characteristic of its dyadic interaction patterns—the *reinforcement phase*.

CONCLUSION

I would be among the first to suggest that the four phases of orientation, conflict, emergence, and reinforcement will not necessarily be present in all task-oriented small groups. Certainly, task groups subject to external legitimate controls would modify the "natural" context characteristic of the groups studied and might consequently affect the groups' interaction processes. Then, too, perma-

nent groups might deviate from the four-phase pattern in subsequent task performances. The nature and extent of such differences point to the need for further research.

But while this four-phase model of "decision emergence" does not pretend to be applicable to all decision-making groups in all situations, the present study does demonstrate that the process of interaction is susceptible to study and analysis apparently free from overt influences of the socio-emotional dimension. This study demonstrates further that the interaction patterns of task behavior follow a consistent pattern of progression across time as groups make decisions despite the presence of social variables idiosyncratic to specific groups. Of course, the relative length of each phase may differ idiosyncratically among groups, but there is some evidence to believe that the proportionate amount of task behavior does not vary significantly with the social structures of groups.²⁰ Thus, inductively applying this model to similarly unstructured decision-making groups seems plausible.

Further research into task behavior utilizing this category system would probably benefit from several revisions of the system. I suggest that the categories of modification (4) and summary (5) be omitted from further applications of this system.²¹ The category system might also differentiate between "bi-valued" attitudes and "neutral" attitudes in the ambiguous (a) category in the system's third dimension. An additional category of "disagreement" might

²⁰ See, for example, R. Victor Harnack, "An Experimental Study of the Effects of Training in the Recognition and Formulation of Goals upon Intra-Group Cooperation," *SM*, XXII (1955), 31-38.

²¹ Subsidiary to the present study was a theory regarding the progressive but largely indirect modification of decision proposals. Since this theory is not strictly within the scope of this report, it is not reported here.

also be added to complement the category of "agreement" (6). One should keep in mind that this category system was not intended to be a system for comprehensive observation of group behavior. It was developed to serve only the limited purpose of observing more closely the interaction patterns of verbal task behavior free from the overt influences of the socio-emotional dimension.

Perhaps the major contribution of this study is its emphasis on the process of interaction; that is, on determining how communication patterns mediate input variables into a group product. This study demonstrates that interaction patterns can be observed directly and the nature of group task completion from the vantage point of verbal behavior can be understood more clearly.